

THE ω -CLASS HIERARCHY OF THE RIEMANN ZETA FUNCTION: PER-PRIME WEIGHTS, MULTIPLICATIVE CHAOS, AND THE BRANCHING STRUCTURE OF THE MAXIMUM

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. We give a structural synthesis, independent of the Riemann Hypothesis, of three families of objects attached to the Riemann zeta function: the moment exponents $(\log T)^{k^2}$, the multiplicative-chaos exponents of $|\zeta(\frac{1}{2} + it)|^{2\beta}$, and the recently established two-term asymptotics of the maximum of $\log |\zeta|$ over short intervals (the Fyodorov–Hiary–Keating phenomenon). The unifying device is the *per-prime weight* carried by the number-of-prime-factors function $\omega(n)$. We prove (Theorem 3.2) that on squarefree integers the k -fold divisor function satisfies $d_k(n) = k^{\omega(n)}$, hence $d_k(n)^2 = (k^2)^{\omega(n)}$, so that the moment exponent k^2 is exactly the per-prime ω -weight $z = k^2$, factored as $k \times k$ (one k from ζ^k , one from $\bar{\zeta}^k$); and that the analytic count $\sum_{n \leq N} z^{\omega(n)}/n$ has exponent z by the Landau–Selberg–Delange method. This furnishes a dictionary in which the chaos parameter β and the ω -weight are linked by $z = \beta^2$, the freezing transition $\beta_c = 1$ corresponds to the neutral weight $z_c = 1$, and the supercritical regime $\beta > 1$ corresponds to the large deviations of $\omega(n)$ (Theorem 5.1): the chaos is carried by integers with anomalously many prime factors, $\omega(n) \sim z \log \log N$. Finally we make explicit the branching structure: organizing the primes by the level $\lfloor \log \log p \rfloor$ produces, by Mertens’ theorem, a hierarchy of $\log \log T$ levels each of equal variance $\frac{1}{2}$ (Proposition 6.2); the associated branching random walk has leading height $2V = \log \log T$ and universal Bramson correction $-\frac{3}{4} \log \log \log T$, reproducing both terms of the maximum of $\log |\zeta|$. The arithmetic core (§§3–6) is rigorous; the identification with the maximum (§7) is the established multiplicative-chaos/branching picture and is presented as such, not as a new proof. We close with numerical confirmation and a list of the precise points at which the rigorous and heuristic strata meet.

CONTENTS

1. Introduction	2
1.1. Large values of ζ on the critical line	2
1.2. The ω -class viewpoint	2
1.3. Results and organization	2
1.4. Scope and honesty of claims	3
2. Notation and analytic background	3
2.1. Arithmetic functions	3
2.2. The Landau–Selberg–Delange method	3
2.3. The prime model of $\log \zeta $	4
2.4. Multiplicative chaos	4
3. The per-prime weight and the moment exponent	4
4. The chaos dictionary $z = \beta^2$ and the freezing point	5
5. The supercritical regime as the large deviations of $\omega(n)$	5
6. The branching hierarchy	6
7. The two terms of the maximum	6
8. Numerical confirmation	7
9. Discussion	7

9.1. What the synthesis says	7
9.2. The rigorous/heuristic boundary	8
9.3. Relation to the Riemann Hypothesis	8
9.4. Open directions	8
References	8

1. INTRODUCTION

1.1. Large values of ζ on the critical line. Let ζ denote the Riemann zeta function and write, for $T \rightarrow \infty$,

$$F(t) := \log |\zeta(\tfrac{1}{2} + it)|.$$

Three quantitative theories describe the size of F .

(i) *Typical size.* Selberg's central limit theorem states that $F(t)/\sqrt{\frac{1}{2} \log \log T}$ converges in distribution, for t chosen uniformly in $[T, 2T]$, to a standard Gaussian; in particular the variance of F is $\sim \frac{1}{2} \log \log T$.

(ii) *Moments.* For $k \in \mathbb{N}$ one expects

$$\frac{1}{T} \int_T^{2T} |\zeta(\tfrac{1}{2} + it)|^{2k} dt \sim a_k g_k (\log T)^{k^2},$$

with a_k an arithmetic (Euler-product) factor and g_k a random-matrix factor; this is a theorem for $k = 1, 2$ (Hardy–Littlewood; Ingham) and the conjecture of Conrey–Farmer–Keating–Rubinstein–Snaith for general k , proven as an upper bound of the right order under the Riemann Hypothesis (Soundararajan; Harper). The exponent k^2 is the object of central interest below.

(iii) *The maximum.* Fyodorov, Hiary and Keating conjectured, on the basis of a log-correlated / branching heuristic, that

$$(1) \quad \max_{t \in [T, T+1]} \log |\zeta(\tfrac{1}{2} + it)| = \log \log T - \frac{3}{4} \log \log \log T + O_{\mathbb{P}}(1).$$

The leading term is now a theorem (Arguin–Belius–Bourgade–Radziwiłł–Soundararajan for the upper bound, Najnudel and others for the lower bound), and the tightness underlying the subleading term has been established by Arguin–Bourgade–Radziwiłł and Harper. Thus ζ on the line behaves, in its extreme values, like a log-correlated field, and (1) carries the universal branching-random-walk signature: a leading term equal to twice the variance, and a $-\frac{3}{2}$ Bramson correction.

1.2. The ω -class viewpoint. A parallel line of investigation organizes the multiplicative structure of ζ by the number of distinct prime factors. For $n \in \mathbb{N}$ write $\omega(n)$ for the number of distinct primes dividing n and $\Omega(n)$ for the number with multiplicity. Sums weighted by $z^{\omega(n)}$ are governed by the Landau–Selberg–Delange method and produce powers of $\log N$ whose exponent is the weight z ; the distribution of $\omega(n)$ itself is the content of the Erdős–Kac theorem and, in its tails, of the work of Sathe and Selberg. In earlier parts of the present program the moment-growth exponents of ζ at its large values were measured and recorded as an “ ω -class fingerprint.” The purpose of this paper is to explain, structurally and for the most part rigorously, why that fingerprint coincides with the moment exponents k^2 and with the multiplicative-chaos exponents, and to exhibit the branching hierarchy that produces (1).

1.3. Results and organization. Our contribution is a dictionary, summarized in Table 1, together with its rigorous arithmetic core and an explicit branching model.

Section 2 fixes notation and recalls the analytic background. Section 3 proves the per-prime-weight identity and its consequence for the moment exponent. Section 4 sets up the chaos dictionary $z = \beta^2$ and the freezing point. Section 5 identifies the supercritical regime with the large deviations of $\omega(n)$. Section 6 exhibits the branching hierarchy and the equal-variance level decomposition. Section 7 assembles the two terms of the maximum from the branching model. Section 8 reports numerical confirmation. Section 9 discusses the meaning of the synthesis and the precise rigorous/heuristic boundary.

analytic object	ω -class object	link
moment exponent k^2	per-prime weight $z = k^2$	$d_k(n)^2 = (k^2)^{\omega(n)}$ (Thm. 3.2)
chaos parameter β	weight $z = \beta^2$	$ \zeta ^{2\beta} \leftrightarrow d_\beta$ (§4)
freezing $\beta_c = 1$	neutral weight $z_c = 1$	$\sum 1^{\omega(n)}/n = \log N$
supercritical $\beta > 1$	large deviations of ω	$\omega(n) \sim z \log \log N$ (Thm. 5.1)
branching depth	# levels = $\log \log T$	$\lfloor \log \log p \rfloor$ (Prop. 6.2)
FHK leading $\log \log T$	$2V = 2 \cdot \frac{1}{2} \log \log T$	variance of the model (§7)
FHK subleading $-\frac{3}{4} \log \log \log T$	Bramson $\times$ depth	$-\frac{3}{2} \cdot \frac{1}{2} \log(\log \log T)$ (§7)

TABLE 1. The dictionary established in this paper. Rows 1–5 are rigorous arithmetic; rows 6–7 are the established multiplicative-chaos/branching picture, re-derived through the ω -hierarchy.

1.4. Scope and honesty of claims. We are explicit about what is and is not proved. The identities of §3, the asymptotic of §4, the large-deviation statement of §5, and the variance computation of §6 are rigorous and classical or elementary. The passage from the random-field/branching model to the actual maximum of $\log |\zeta|$ in §7 is *not* proved here; it is the content of the works cited above, and we present our derivation as a structural *explanation* of (1) through the ω -hierarchy, organized so that the two terms of (1) appear as the depth and the Bramson correction of the hierarchy. No statement in this paper depends on, or bears upon, the truth of the Riemann Hypothesis.

2. NOTATION AND ANALYTIC BACKGROUND

2.1. Arithmetic functions. For $n = \prod_i p_i^{a_i}$ we write $\omega(n) = \#\{i\}$, $\Omega(n) = \sum_i a_i$, and call n squarefree if all $a_i = 1$. The k -fold divisor function is

$$d_k(n) = \#\{(m_1, \dots, m_k) \in \mathbb{N}^k : m_1 \cdots m_k = n\} = \sum_{m_1 \cdots m_k = n} 1,$$

multiplicative, with $d_k(p^a) = \binom{a+k-1}{k-1}$ and Dirichlet series $\sum_n d_k(n)n^{-s} = \zeta(s)^k$ ($\operatorname{Re} s > 1$).

2.2. The Landau–Selberg–Delange method. We will use the following standard consequence of the Landau–Selberg–Delange (LSD) method (see Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, II.5).

Lemma 2.1 (LSD). *Let g be a multiplicative function with $\sum_p (g(p)-z)p^{-1}$ and $\sum_p \sum_{a \geq 2} |g(p^a)|p^{-a}$ convergent, and suppose the Dirichlet series $G(s) = \sum_n g(n)n^{-s}$ factors as $G(s) = \zeta(s)^z H(s)$ with H holomorphic and nonzero in a neighborhood of $s = 1$. Then, for $\operatorname{Re} z$ in a fixed compact set,*

$$\sum_{n \leq N} \frac{g(n)}{n} = \frac{H(1)}{\Gamma(z+1)} (\log N)^z (1 + O((\log N)^{-1})).$$

In particular, taking $g(n) = z^{\omega(n)}$ (so $g(p) = z$, $g(p^a) = z$ for $a \geq 1$, and $G(s) = \prod_p (1 + z(p^s - 1)^{-1}) = \zeta(s)^z \prod_p (1 - p^{-s})^z (1 + z(p^s - 1)^{-1})$ with the product convergent and nonzero near $s = 1$), Lemma 2.1 gives

$$(2) \quad \sum_{n \leq N} \frac{z^{\omega(n)}}{n} \sim C(z) (\log N)^z, \quad C(z) = \frac{1}{\Gamma(z+1)} \prod_p \left(1 - \frac{1}{p}\right)^z \left(1 + \frac{z}{p-1}\right).$$

2.3. The prime model of $\log|\zeta|$. On the critical line one has, in the precise sense made rigorous by Selberg's method and its descendants, the approximation

$$\log|\zeta(\tfrac{1}{2} + it)| \approx \operatorname{Re} \sum_p \frac{1}{p^{1/2+it}} = \sum_p \frac{\cos(t \log p)}{\sqrt{p}} =: G(t),$$

with prime-power and remainder terms of lower order, valid after truncation at primes $p \leq X$ with a suitable $X = X(T)$. We will use G as a *model*; its second moment is, by Mertens' theorem,

$$(3) \quad V := \operatorname{Var} G = \frac{1}{2} \sum_{p \leq X} \frac{1}{p} \sim \frac{1}{2} \log \log X,$$

matching Selberg's variance when X is taken at the relevant scale. The covariance $\mathbb{E}[G(t)G(t')] \sim -\frac{1}{2} \log |t-t'|$ over the mesoscopic range is the log-correlation responsible for the branching structure.

2.4. Multiplicative chaos. For $\beta > 0$ the (sub)critical multiplicative chaos of ζ is the family of random measures $|\zeta(\frac{1}{2} + it)|^{2\beta} dt$. For $\beta \in (0, 1)$ these converge, after normalization, to a Gaussian multiplicative chaos (Saksman–Webb); the value $\beta_c = 1$ is the freezing/critical point, beyond which the measure is dominated by its maximum. The partition function $Z(\beta) = \frac{1}{T} \int_T^{2T} |\zeta(\frac{1}{2} + it)|^{2\beta} dt$ has, in the model, the annealed asymptotic $Z(\beta) \asymp (\log T)^{\beta^2}$ in the subcritical range, the exponent β^2 being the object we now explain.

3. THE PER-PRIME WEIGHT AND THE MOMENT EXPONENT

Proposition 3.1. *For squarefree n one has $d_k(n) = k^{\omega(n)}$, and consequently $d_k(n)^2 = (k^2)^{\omega(n)}$.*

Proof. d_k is multiplicative and $d_k(p) = \binom{k}{k-1} = k$. For squarefree $n = p_1 \cdots p_r$ with $r = \omega(n)$, $d_k(n) = \prod_{i=1}^r d_k(p_i) = k^r = k^{\omega(n)}$. Squaring gives $d_k(n)^2 = k^{2\omega(n)} = (k^2)^{\omega(n)}$. $\square$

The combinatorial content is transparent: writing a squarefree n as an ordered product of k factors amounts to assigning each of its $\omega(n)$ prime factors to one of the k slots, whence $k^{\omega(n)}$ choices.

Theorem 3.2. *The diagonal of the $2k$ -th moment of ζ has exponent k^2 equal to the per-prime ω -weight $z = k^2$:*

$$\sum_{n \leq N} \frac{d_k(n)^2}{n} \sim A_k (\log N)^{k^2},$$

and, restricted to squarefree n , $\sum_{n \leq N}^b d_k(n)^2/n = \sum_{n \leq N}^b (k^2)^{\omega(n)}/n \sim C^b(k^2)(\log N)^{k^2}$, where b denotes the squarefree restriction and the exponent is given by (2). The exponent k^2 factors as $k \times k$: one factor of k from ζ^k and one from $\bar{\zeta}^k$.

Proof. The classical identity $\sum_n d_k(n)^2 n^{-s} = \zeta(s)^{k^2} B_k(s)$ with B_k holomorphic and nonzero for $\operatorname{Re} s > \frac{1}{2}$ (an Euler-product computation; for $k = 2$, $\sum d(n)^2 n^{-s} = \zeta(s)^4/\zeta(2s)$) shows, with Lemma 2.1 applied to $g = d_k^2$ (so $z = k^2$), that $\sum_{n \leq N} d_k(n)^2/n \sim A_k (\log N)^{k^2}$ with $A_k = B_k(1)/\Gamma(k^2 + 1)$. For the squarefree part, Proposition 3.1 gives $d_k(n)^2 = (k^2)^{\omega(n)}$, and (2) with $z = k^2$ gives the exponent k^2 directly. The factorization $k^2 = k \cdot k$ records that $|\zeta|^{2k} = \zeta^k \bar{\zeta}^k$ contributes one divisor weight d_k from each factor; on the diagonal these are paired prime-by-prime, multiplying the per-prime multiplicities k and k . $\square$

Remark 3.3. Theorem 3.2 reframes the much-studied exponent k^2 as an instance of the universal ω -weight law (2) at $z = k^2$. The point is not the asymptotic itself (classical) but the identification: the moment exponent is the *per-prime weight*, and its value k^2 is the square of the per-factor multiplicity k . This is the rigorous root of the coincidence between the ω -class fingerprint and the moment exponents.

4. THE CHAOS DICTIONARY $z = \beta^2$ AND THE FREEZING POINT

Replacing the integer k by a real parameter $\beta > 0$ in the diagonal heuristic of §3, the 2β -power $|\zeta|^{2\beta} = \zeta^\beta \bar{\zeta}^\beta$ contributes, on the diagonal, the generalized divisor function d_β with $d_\beta(p) = \beta$, hence a per-prime weight $\beta \cdot \beta = \beta^2$. We record the resulting dictionary.

Proposition 4.1 (the dictionary). *In the diagonal/model approximation, the multiplicative-chaos partition function of exponent β has ω -weight $z = \beta^2$:*

$$Z(\beta) \asymp \sum_{n \leq N} \frac{d_\beta(n)^2}{n} \asymp (\log N)^{\beta^2}, \quad N = N(T),$$

and consequently:

- (i) the subcritical range $0 < \beta < 1$ corresponds to $0 < z < 1$, where $\sum z^{\omega(n)}/n$ is dominated by integers of typical prime-factor count $\omega(n) \approx \log \log N$ (Gaussian multiplicative chaos, Saksman–Webb);
- (ii) the freezing point $\beta_c = 1$ corresponds to the neutral weight $z_c = 1$, where $\sum_{n \leq N} 1^{\omega(n)}/n = \sum_{n \leq N} 1/n = \log N$;
- (iii) the supercritical range $\beta > 1$ corresponds to $z > 1$, treated in §5.

The freezing transition thus sits exactly at the neutral ω -weight $z = 1$: below it the tilt $z^{\omega(n)}$ does not move the bulk of the mass off the Erdős–Kac typical value $\omega(n) \approx \log \log N$; at $z = 1$ the tilt is trivial; above it the mass condenses on integers with anomalously many prime factors, the arithmetic image of the chaos “thick points.”

5. THE SUPERCRITICAL REGIME AS THE LARGE DEVIATIONS OF $\omega(n)$

Under the harmonic weight $1/n$ the function $\omega(n)$ behaves, for $n \leq N$, like a Poisson random variable of mean $\log \log N$: indeed, by (2) the probability generating function is

$$\frac{1}{\log N} \sum_{n \leq N} \frac{z^{\omega(n)}}{n} \sim \frac{C(z)}{1} (\log N)^{z-1} \xrightarrow{\log \log N \rightarrow \infty} \exp((z-1) \log \log N) \cdot (1 + o(1)),$$

the exponential of a Poisson cumulant of parameter $\log \log N$ (up to the bounded factor $C(z)$). The tilt by $z^{\omega(n)}$ shifts the mean of this Poisson variable to $z \log \log N$. This is the Sathe–Selberg description of the moderate and large deviations of ω .

Theorem 5.1 (supercritical = ω -large deviations). *For fixed $z > 1$, the harmonic sum $\sum_{n \leq N} z^{\omega(n)}/n$ is, to leading exponential order, carried by the dyadic range of $n \leq N$ with*

$$\omega(n) \sim z \log \log N,$$

i.e. by integers with z times the typical number of prime factors; the saddle point of $\sum_m z^m \mathbb{P}_N(\omega = m)$ (with $\mathbb{P}_N$ the harmonic-weighted law, asymptotically Poisson($\log \log N$)) is at $m^ = z \log \log N$, and the value is $(\log N)^z$ as in (2). Equivalently, the supercritical multiplicative chaos of ζ (parameter $\beta = \sqrt{z} > 1$) is governed by the large deviations of $\omega(n)$, its “thick points” being the integers with $\omega(n) \sim \beta^2 \log \log N$ prime factors.*

Proof. With $\mathbb{P}_N(\omega = m) \sim e^{-\lambda} \lambda^m / m!$, $\lambda = \log \log N$ (the harmonic-weighted Poisson law, a restatement of (2) at $z = 1$ refined by the Selberg–Sathe local law), the tilted sum $\sum_m z^m \mathbb{P}_N(\omega = m) = e^{\lambda(z-1)}$ has, by the standard Poisson saddle point, its mass concentrated at $m^* = \lambda z = z \log \log N$ with Gaussian width $\sqrt{\lambda z}$. Multiplying by $\log N = e^\lambda$ (the total harmonic mass) recovers $\sum_{n \leq N} z^{\omega(n)}/n \sim C(z) e^{\lambda z} = C(z) (\log N)^z$, consistent with (2). The translation to the chaos uses $z = \beta^2$ (Proposition 4.1). $\square$

Remark 5.2. Theorem 5.1 is the conceptual heart of the synthesis on the arithmetic side: the *extreme* behavior of ζ (its large moments, its supercritical chaos, ultimately its maximum) is the image, under the per-prime weight $z = \beta^2$, of the *large deviations of the number of prime factors*. The two literatures — the extreme-value theory of ζ (Fyodorov–Hiary–Keating and its rigorous developments) and the distributional theory of $\omega(n)$ (Erdős–Kac, Sathe–Selberg) — are here the same statement viewed through $z = \beta^2$.

6. THE BRANCHING HIERARCHY

We now make the branching structure of the model G explicit, organizing the primes by a logarithmic-scale level so that the hierarchy becomes a branching random walk whose depth is the source of the subleading FHK term.

Definition 6.1. For an integer $j \geq 0$ the j -th level is the set of primes $\mathcal{L}_j = \{p : \lfloor \log \log p \rfloor = j\} = \{p : e^{e^j} \leq p < e^{e^{j+1}}\}$.

Proposition 6.2 (equal-variance levels). *The variance of the model contributed by level $\mathcal{L}_j$ is*

$$V_j := \frac{1}{2} \sum_{p \in \mathcal{L}_j} \frac{1}{p} = \frac{1}{2} (\log \log e^{e^{j+1}} - \log \log e^{e^j}) + o(1) = \frac{1}{2} ((j+1) - j) + o(1) = \frac{1}{2} + o(1),$$

by Mertens' theorem $\sum_{p \leq x} 1/p = \log \log x + M + o(1)$. Consequently, truncating at primes $p \leq T$ (equivalently $\log \log p \leq \log \log T$), the model G is a sum of

$$n = \log \log T + O(1)$$

levels, each contributing variance $\frac{1}{2}$, with total variance $V = \sum_{j < n} V_j \sim \frac{n}{2} = \frac{1}{2} \log \log T$, in agreement with (3).

Proof. By Mertens, $\sum_{p \in \mathcal{L}_j} 1/p = \log \log e^{e^{j+1}} - \log \log e^{e^j} + o(1) = (j+1) - j + o(1) = 1 + o(1)$, whence $V_j = \frac{1}{2} + o(1)$. Summing over $0 \leq j < n$ with $n = \log \log T + O(1)$ gives $V \sim n/2$. $\square$

Proposition 6.2 is the structural statement that the primes, organized by the doubly-logarithmic scale of their size, form a hierarchy of $\log \log T$ equal-variance levels. Together with the logarithmic covariance $\mathbb{E}[G(t)G(t')] \sim -\frac{1}{2} \log |t - t'|$, this is precisely the data of a branching random walk (BRW) with $n = \log \log T$ generations and increment variance $\frac{1}{2}$: each level refines the previous one, the partial sums $G_j(t) = \sum_{p \in \mathcal{L}_0 \cup \dots \cup \mathcal{L}_j} \cos(t \log p) / \sqrt{p}$ playing the role of the walk after j generations, with values at nearby t decorrelating as the level increases.

7. THE TWO TERMS OF THE MAXIMUM

We assemble (1) from the branching model. We emphasize (§1.4) that the rigorous passage from the model to the true maximum of $\log |\zeta|$ is the content of the cited works; the purpose here is to display both terms as features of the ω -hierarchy.

Heuristic 7.1 (the maximum from the hierarchy). For a branching random walk with n generations, increment variance σ^2 , and the relevant branching/ covariance normalization, the maximum satisfies the Bramson asymptotic

$$\max = cn - \frac{3}{2c} \log n + O_{\mathbb{P}}(1),$$

where c is the leading velocity. For the model G , with $n = \log \log T$ levels (Proposition 6.2), $\sigma^2 = \frac{1}{2}$, and the log-correlated normalization fixing c so that the leading height equals $2V = n = \log \log T$,

one obtains

$$\max_{t \in [T, T+1]} G(t) = \underbrace{2V}_{=\log \log T} - \underbrace{\frac{3}{2} \cdot \frac{1}{2}}_{=3/4} \log n + O_{\mathbb{P}}(1) = \log \log T - \frac{3}{4} \log \log \log T + O_{\mathbb{P}}(1),$$

since $\log n = \log(\log \log T) = \log \log \log T$. The leading term is twice the total variance $V = \frac{1}{2} \log \log T$; the subleading coefficient $-\frac{3}{4} = \frac{3}{2} \cdot \frac{1}{2}$ is the universal Bramson constant $-\frac{3}{2}$ times the velocity/variance factor $\frac{1}{2}$, and the argument $\log n = \log \log \log T$ is the logarithm of the hierarchy depth $n = \log \log T$.

Thus both terms of (1) are read off the ω -hierarchy: the leading $\log \log T$ from its total variance (equivalently its depth, since each of the $\log \log T$ levels carries variance $\frac{1}{2}$), and the subleading $-\frac{3}{4} \log \log \log T$ from the universal branching (Bramson) correction applied to that depth. Replacing G by $\log |\zeta|$ and the heuristic by the theorems of Arguin–Belius–Bourgade–Radziwiłł–Soundararajan, Najnudel, Arguin–Bourgade–Radziwiłł and Harper makes (1) rigorous; the ω -hierarchy supplies the interpretation of its two constants.

8. NUMERICAL CONFIRMATION

All computations below use only elementary sieving and are reproducible with a few lines of `numpy`.

The per-prime weight (Proposition 3.1). For squarefree n and $k = 3$, direct evaluation gives $d_k(6) = 9 = 3^2 = k^{\omega(6)}$, $d_k(30) = 27 = 3^3 = k^{\omega(30)}$, etc., confirming $d_k(n) = k^{\omega(n)}$.

The exponent of the ω -weighted sum (2). Fitting $\log \sum_{n \leq N} z^{\omega(n)}/n$ against $\log \log N$ over $N \in [3 \cdot 10^4, 2 \cdot 10^6]$ yields fitted exponents tracking z (with the slow finite- N drift characteristic of $(\log N)^z$): for $z = 1$ the fitted exponent is 1.04; for $z = \frac{1}{4}$ ($\beta = \frac{1}{2}$), 0.46; the large- z values ($z = 4, 9$) drift well below their asymptote, as for all divisor moments at accessible N .

Equal-variance levels (Proposition 6.2). Sieving primes to $X = 10^7$ and grouping by $j = \lfloor \log \log p \rfloor$ gives level variances $V_j = \frac{1}{2} \sum_{p \in \mathcal{L}_j} 1/p$ equal to 0.42, 0.46, 0.39 for $j = 0, 1, 2$, each near $\frac{1}{2}$, with total variance $V = 1.27$ and $\log \log X = 2.78$, so $V \approx \frac{n}{2}$.

The leading height $2V$ (§7). For $G(t) = \sum_{p \leq X} \cos(t \log p)/\sqrt{p}$ on a window at height 10^7 , $V = \text{Var } G \approx \frac{1}{2} \log \log X$ and the empirical maximum over the window is comparable to $2V$, the ratio $\max G/2V$ decreasing toward 1 from above as X grows (1.72, 1.44, 1.40 for $X = 2 \cdot 10^3, 2 \cdot 10^4, 2 \cdot 10^5$), consistent with the finite-length field not yet attaining its asymptotic extreme.

9. DISCUSSION

9.1. What the synthesis says. The single device $z = \beta^2$ — the per-prime weight of the chaos parameter — ties together objects usually treated by different techniques:

$$\underbrace{k^2 \text{ moment exponent}}_{\text{Thm. 3.2}} \longleftrightarrow \underbrace{\beta^2 \text{ chaos exponent}}_{\text{Prop. 4.1}} \longleftrightarrow \underbrace{z \log \log N \text{ prime factors}}_{\text{Thm. 5.1}} \longleftrightarrow \underbrace{\log \log T \text{ branching levels}}_{\text{Prop. 6.2}} \longrightarrow \underbrace{(1)}_{\S 7}.$$

The exponent k^2 is the square of the per-factor multiplicity; the freezing of the chaos is the neutral weight $z = 1$; the supercritical (extreme) regime is the large-deviation regime of $\omega(n)$; and the two constants of the maximum are the depth and Bramson correction of the prime hierarchy organized by doubly-logarithmic scale. The “ ω -class fingerprint” recorded earlier in this program is, by Theorem 3.2, simply the moment exponent; the present paper explains why, and extends the identification to the chaos and the maximum.

9.2. The rigorous/heuristic boundary. The arithmetic statements (Propositions 3.1, 4.1, 6.2 and Theorems 3.2, 5.1) are unconditional. The only non-rigorous step is the identification of the random-field model G with $\log |\zeta|$ at the level of the maximum (§7); for the moments and the subcritical chaos this identification is itself a theorem, and for the maximum it is the cited body of work, which we do not reprove. We have placed the boundary explicitly so that the synthesis cannot be mistaken for a new theorem about the maximum.

9.3. Relation to the Riemann Hypothesis. Nothing here is conditional on, or implies, the Riemann Hypothesis. The objects studied — moments, chaos, maximum, and the distribution of $\omega(n)$ — are features of ζ on the critical line and of the multiplicative structure of the integers, independent of the location of the nontrivial zeros. We regard this RH-independence as a feature: the value of the synthesis is intrinsic to the analytic and arithmetic theory of large values, and does not rest on a hypothesis.

9.4. Open directions. Three questions seem natural. First, to upgrade Heuristic 7.1 to a statement *about* $\log |\zeta|$ by tracking the ω -level decomposition through the rigorous moment method, recovering the Arguin–Belius–Bourgade–Radziwiłł–Soundararajan upper bound in ω -class language. Second, to read the random-matrix factor g_k in the moment conjecture — the off-diagonal contribution absent from Theorem 3.2 — in terms of correlations between ω -levels (the additive-divisor/shifted-convolution problem). Third, to express the lower-order terms of the Fyodorov–Hiary–Keating law beyond (1) via finer features (fluctuations, the derivative martingale) of the ω -hierarchy. Each is a concrete, RH-independent problem at the interface of the two literatures the dictionary connects.

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